SHASHWAT YADAV

Engineering Portfolio

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Introduction

I am an aerospace engineering student with a focus on dynamics, control systems, and aerodynamics, drawn to problems at the edge of what current engineering can reliably solve. My work spans active UAV controls research, computational aerodynamic design for motorsport, and a consistent track record of taking projects from concept through simulation to physical hardware. I am drawn to organizations pushing the limits of what flight, propulsion, and high-speed vehicles can do, whether in space, on a racetrack, or anywhere in between. This portfolio documents the technical depth and progression behind each project I have undertaken.

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Robust Control Framework for Retractable Tether Multirotor UAVs (RTMUAVs)

Dynamics and Control Lab at University of South Carolina
January 2025 - Present

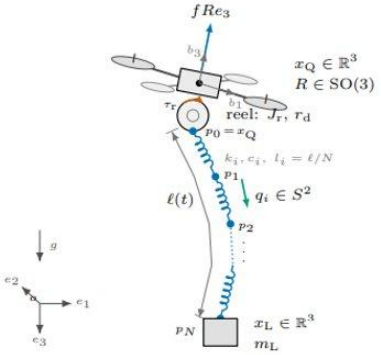
Tethered UAV (TMUAV) systems offer compelling advantages in persistent aerial operations, especially retractable tethers (RTMUAV), however they introduce a fundamentally difficult controls problem: the system must remain stable as the tether dynamically transitions between a slack (unconstrained) and taut (tension-loaded) state. Currently, a controller framework capable of handling both of these states and employing RTMUAV use in high risk situations does not exist. Thus the goal of this project is to build a controller that is robust against in these conditions allowing for expanded use of RTMUAVs.

UAV Dynamics & System Modeling

- Developed a full coupled dynamic model of the UAV, tether, and payload as a unified system — capturing the nonlinear interactions between all three bodies across both tether states
- Tested and successfully controller the UAV against disturbances triggering slack state mid-flight

Tether Modeling

- Modeled the tether under both slack and taut conditions, incorporating state-switching logic with adaptive gains to handle the transition dynamics that destabilize conventional controllers
- Used spring damper model to accurately predict and model tether tension and dynamics
- Multilink model in order to more accurately model tether



Goals

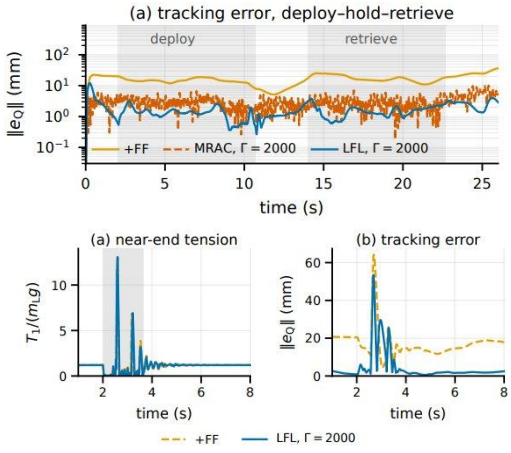
- Properly address tether dynamics during reeling movements
- Maintain stability in cases of slack to taut re-engagements and wind or other disturbances
- Show tracking errors are within bounded errors for stability throughout mission horizon
- Incorporate realistic uncertainties and disturbances to validate controller robustness

Disturbance Modeling

- Realistic disturbances into the modeling and tests
- 8-tone gust field
- Sensor noise included in tension feedback term

Low Frequency Learning (LFL)

- Filter through high frequency noise by comparing
- it to a weight estimate, eliminating leakage bias
- MRAC - updates weights without filter
- Tracking ability vs MRAC
  - Tracked much more precisely
  - Required less control rate effort from the UAV
  - Same learning rate as MRAC
- Thus proven to be an effective addition to the controller in order to ensure more effective response



Awards

- NASA SC Space Grant Consortium Undergraduate Research Award — 2026
- 2nd Place · IMECE 2025 ASME Undergraduate Expo
- Magellan Mini Undergraduate Research Grant — 2025

Presentations

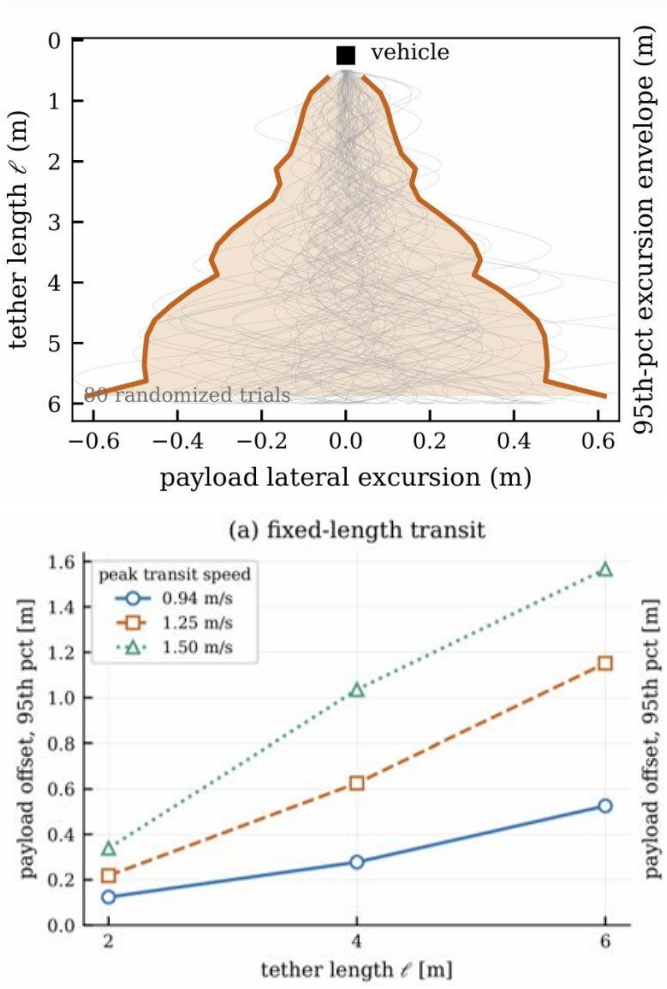
- IMECE 2025 ASME Undergraduate Expo — Poster
- Discover USC 2026 — Poster

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Certified Envelopes for RTMUAV operation

- Maneuver envelopes to replace fixed safety margins with maneuver-specific, validated clearance requirements
- Ran randomized simulations and extracted 95th-percentile excursion envelope
- Tested with tether lengths from 2 to 6 meters
- Tested and compared results from 2D and 3D simulations
- Hover and Reel
  - Longer tether and larger disturbance generally increase required clearance
  - Full 3D validation shows the planar model needs a fidelity correction
- Transit
  - Envelope increases strongly with both speed and tether length
  - Planning can trade transit speed against required clearance
- Swing Arrest
  - Peak excursion is similar in planar and 3D (mostly initial swing)
  - Gain sweep showed tuning helps, but does not eliminate the discrepancy
- In some runs there was a very large cone for operations at tether length of 2 meters

Reeling Platform

- Novel reeling system that weighs under 500-800 g and tracks tension and directions of tether
- Previously only done with ground-linked systems, which have heavy infrastructure, thus scaling down model was a challenge
- A two-axis gimbal with rotary sensors
- measures tether direction
- A reel-rotation sensor tracks tether
- deployment and retrieval
- The package provides the near-end
- tension value needed for onboard
- controller calculations.
- KEY REQUIREMENTS
  - Lightweight module and sensor package
  - Required payload/load capability
  - Commanded reel speed within the available power budget
  - Onboard tension and direction measurements for flight testing



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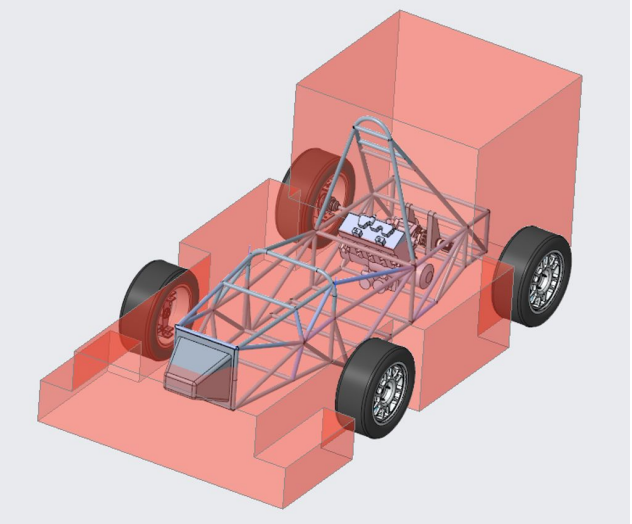
Aerodynamics Team Lead for Society of Automotive Engineers (SAE)

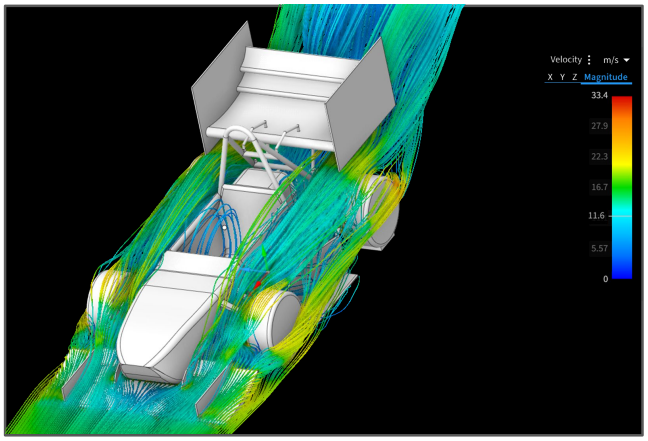
University of South Carolina
May 2025 - Present

After re-establishing the University of South Carolina's Formula SAE program from scratch with the help of the other team leads, I took on the role of Aerodynamics Team Lead, leading the group into FSAE Michigan 2027. In this process it involved procuring all of the materials and tools needed to complete our work, talking with college heads to gain access to funding and space to construct car, and extensive recruitment opportunities to get as many people helping in getting the car finished as possible.

Aerodynamic Package Development

- Leading design of the full external aerodynamic package including front wing, rear wing, nose, and sidepods
- Gained over 100+ hours of CREO Parametric Experience and over 20+ hours of CATIA 3dexperience
- Performed surface modeling in CATIA 3DExperience to explore and optimize aerodynamic bodywork geometry prior to committing to CAD-defined surfaces
- Built full vehicle CAD assemblies in CREO Parametric, integrating aerodynamic components with chassis and suspension geometry to ensure packaging compatibility





Computational Fluid Dynamics (CFD)

- Gained experience in ANSYS softwares such as ANSYS Discovery and ANSYS Fluent
- Executed 20+ ANSYS Fluent CFD simulations across iterative bodywork configurations
- Led study sessions into different turbulence models that are available and their uses
- Learned about result analysis of vorticity, total pressure loss, and different flow profiles

Manufacturing

- Gained foundational exposure to composite manufacturing techniques including carbon fiber wet layup, autoclave processing, and fiberglass fabrication
- Looked into pros and cons of various materials in body fabrication such as aluminum, fibreglass, and carbon fiber to optimize material choice given limited cost budget
- Assembled and mounted components on a go-kart development mule to build hands-on mechanical familiarity ahead of full car assembly

Team Leadership & Program Development

- Re-established the USC FSAE program from the ground up, recruiting, organizing, and leading a team of 8 engineers for aerodynamics
- Developed onboarding curriculum to train new members on ANSYS Fluent, building team-wide simulation competency from scratch
- Helped secured industry sponsorships to fund team operations, tools and components

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Scramjet Supersonic Isolator Unstart and Shock Train Transient Simulation

Personal Project

October 2026 - Present

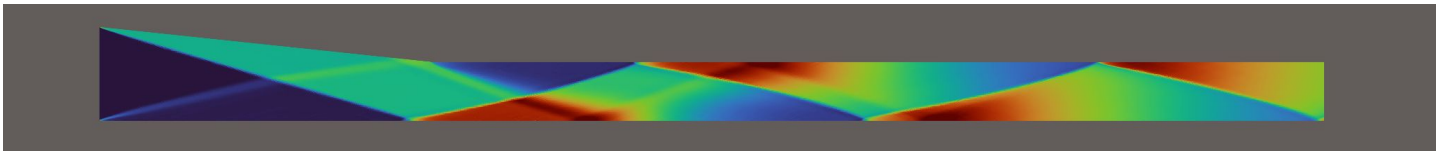
Supersonic inlets and isolators are critical components in scramjet propulsion systems. However, they face a severe operational limitation: back-pressure spikes can cause the internal shock train to migrate upstream, leading to complete inlet "unstart" and catastrophic thrust collapse. While steady-state behavior is well-documented, capturing the transient relaxation response and shock-train anchoring mechanics under specific forcing functions remains a complex computational challenge. Thus, the goal of this study is to model and analyze the transient unstart mechanics of a supersonic isolator under controlled back-pressure and inflow forcing, including hysteresis and vehicle acceleration, to evaluate dynamic stability boundaries.

Compressible Flow and Solver Setup (ANSYS Fluent)

- Set up a compressible Navier-Stokes solver with the energy equation in ANSYS Fluent to resolve severe thermal gradients and shock-boundary layer interactions (SBLI)
- Modeled air as an ideal gas with temperature-dependent molecular viscosity to capture temperature and pressure changes through supersonic flow
- Applied the k-omega SST turbulence model to capture boundary layer separation, large recirculation zones, and the steep adverse pressure gradients of internal shock trains
- Used fine time increments in the transient solver to resolve high-frequency pressure waves while minimizing numerical diffusion

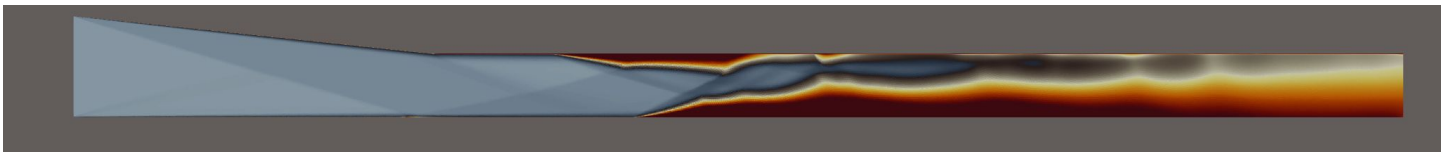
Steady-State Baseline (Mach 4.9) (with your Steady_State.png)

- Set up the Mach 4.9 inflow and isolator geometry following the UT Austin inlet-isolator experiment, with a freestream static pressure of about 5 kPa
- The converged solution shows an oblique shock from the inlet ramp reflecting between the walls and forming a shock train down the duct
- Used this solution as the initial condition for all transient cases
- Quantitative comparison with the experimental wall pressure data is in progress



Linear Ramp Transient Simulation

- Starting from the steady baseline, raised the outlet back pressure linearly to a 200 kPa cap at 2e8 Pa/s, then held it until the flow reached a new steady state
- The ramp simulates progressive downstream throttling and drives the shock train upstream, and the hold shows where the shock system settles
- The shock structure moves upstream and a growing separated region constricts the flow, affecting mass flow rate into the combustor



Hysteresis Study (In Progress)

- Ramping the back pressure up to the cap and then back down to the baseline to see whether the shock train returns to its original position
- Comparing shock-train location on the way up and down at the same back pressure to quantify hysteresis and relaxation time scales
- Aims to map restart and recovery limits after unstart

Accelerating Inflow Study (In Progress)

- Adding a time-varying inflow Mach number to the back-pressure forcing to represent a vehicle accelerating while combustor pressure rises
- Examines how acceleration rate changes shock-train position and the back pressure at which unstart occurs

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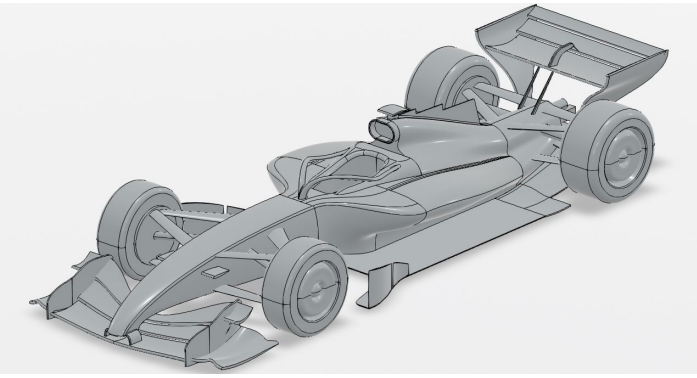
CFD Study of Crosswind/Yaw Effect on Aerodynamics of Formula 1 Cars

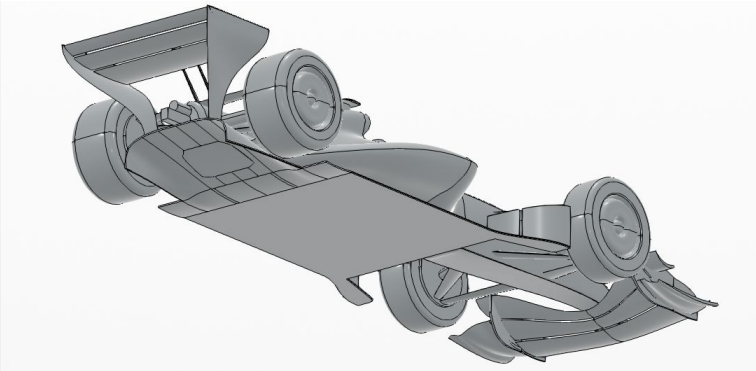
Personal Project
June 2026 - October 2026

Formula 1 is a sport that is known for its engineering challenges and for maximizing the performance out of designs even for advantages of tenths of a second. Working in this kind of environment that I would love to be a part of. For this reason, I chose to pursue a personal summer project that would help me more deeply explore the aerodynamics of Formula 1 cars.

Formula 1 Type Car Development

- Based off of 2026 F1 rules, with some lenient constraints
- Focused on downforce and drag considerations for the underbody, front wing, and rear wing to ensure proper model
- Used CATIA Generative Shape Design to develop extensive and precise CAD model for the car
- Compared simple design with other teams to ensure proper modeling

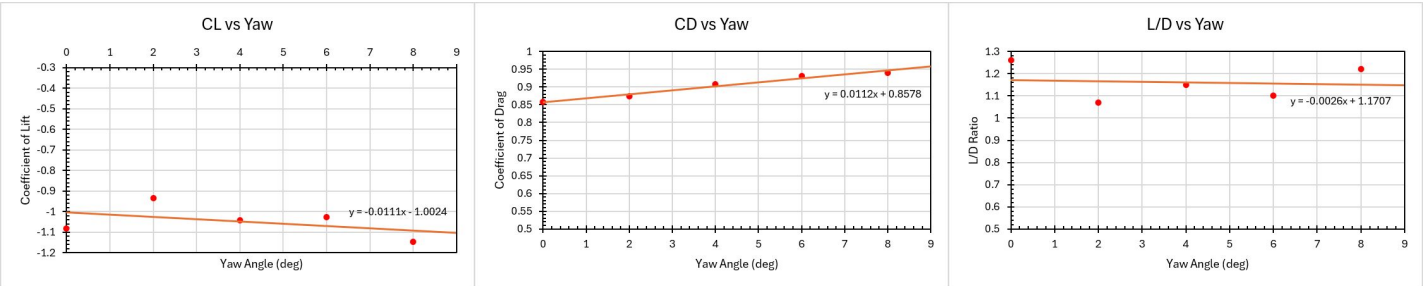




Computational Fluid Dynamics (CFD) Setup

- All results simulated through OpenFOAM
- Formula 1 Car moving at 80000 mm/s in the negative y-direction (thus airflow is 80000 mm/s in the positive y-direction)
    - Kept constant magnitude of velocity in order to conserve dynamic pressure through the study
    - Ground velocity modeled only in y-velocity (only matching velocity of the car)
- Rotating MRFs around all four wheels
- Inlet and Exhaust not modeled, but kept enclosed for lack of engine data
- kOmegaSST turbulence model to properly catch the vortex development, flow separation and viscous effects that would change from yawing winds

General Results (testing results images on next page)



- CL and CD change proportionally with CD increasing slightly faster
- This effect causes a slight drift of the L/D ratio downward
- Conclusion: generally, up to 8 degree crosswinds have no effect of overall L/D ratios of the car, only effects on yawing moments/forces and aerodynamic center

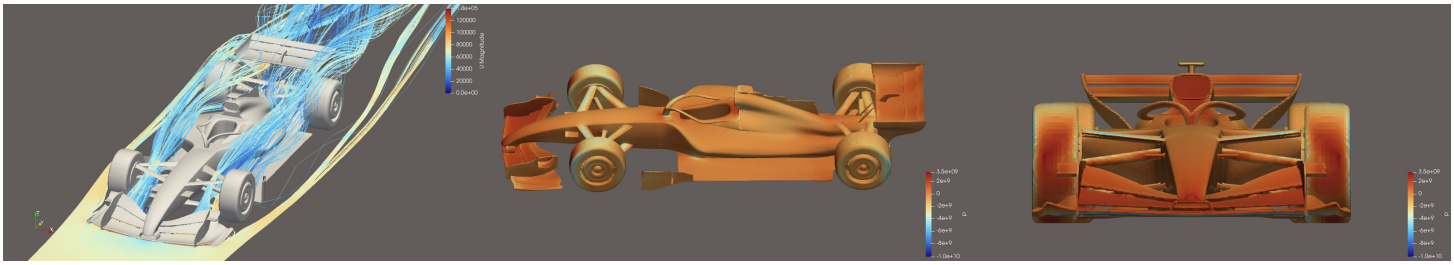
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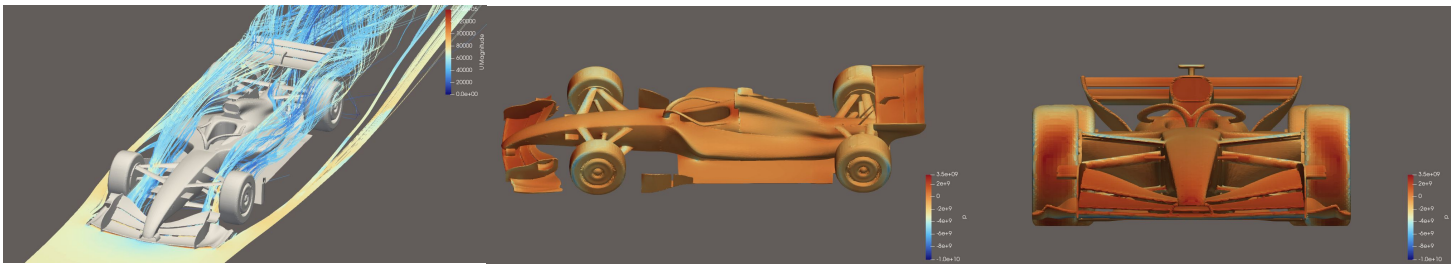
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June 2026 - October 2026

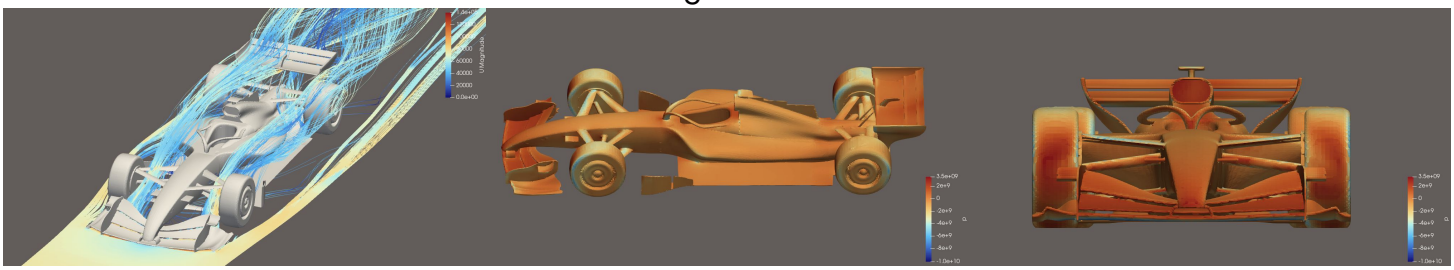
0 degree Results (Baseline)



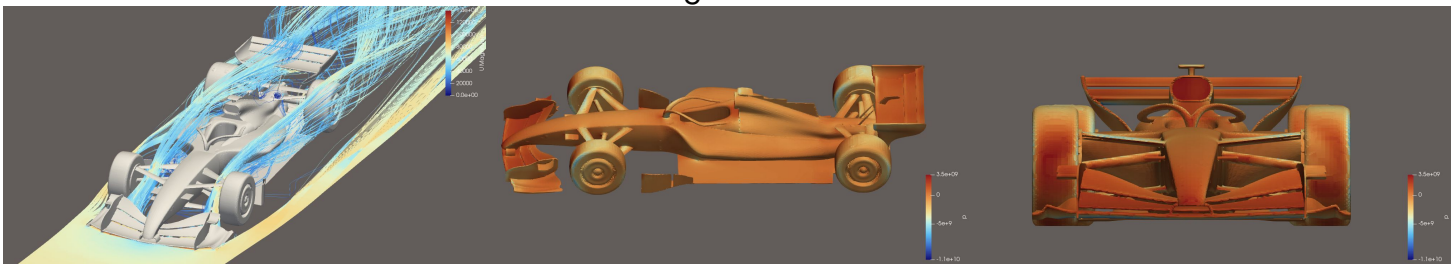
2 degree Results



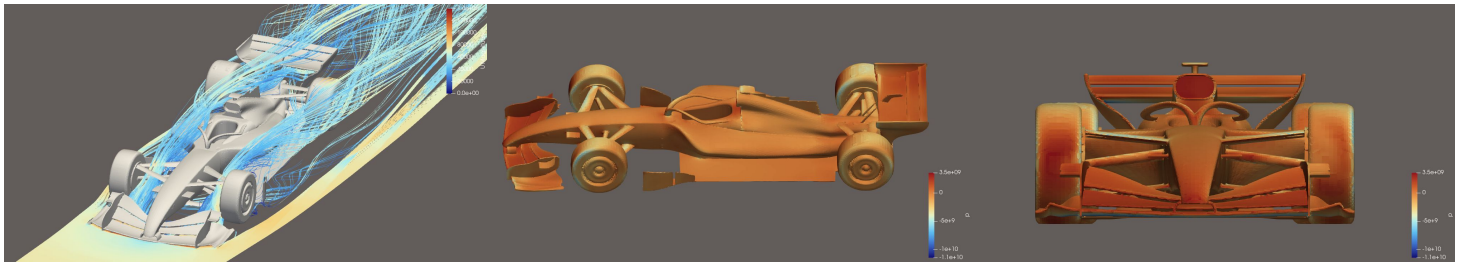
4 degree Results



6 degree Results



8 degree Results



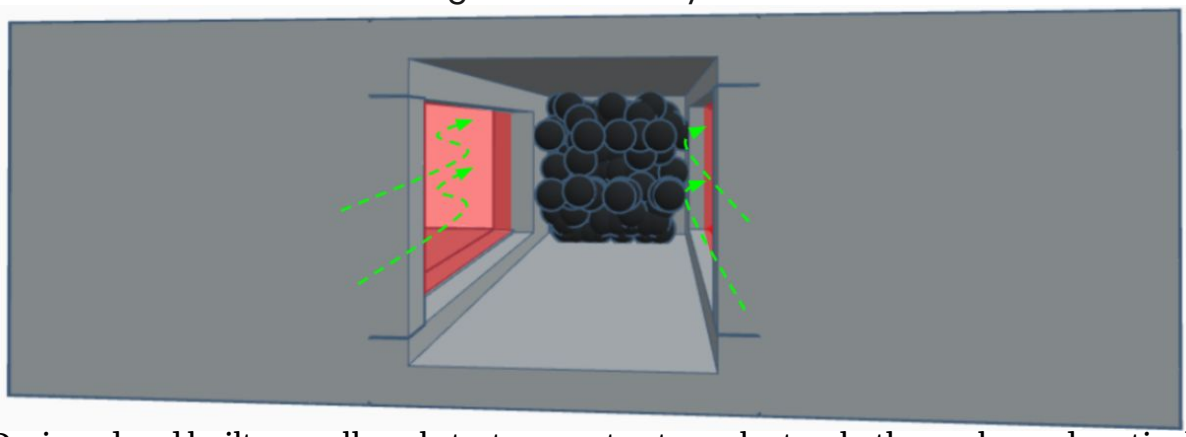
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EARLY RESEARCH

Exploring a Design of a More Efficient System to Remove Soot Load from Diesel Particulate Filters (DPFs)

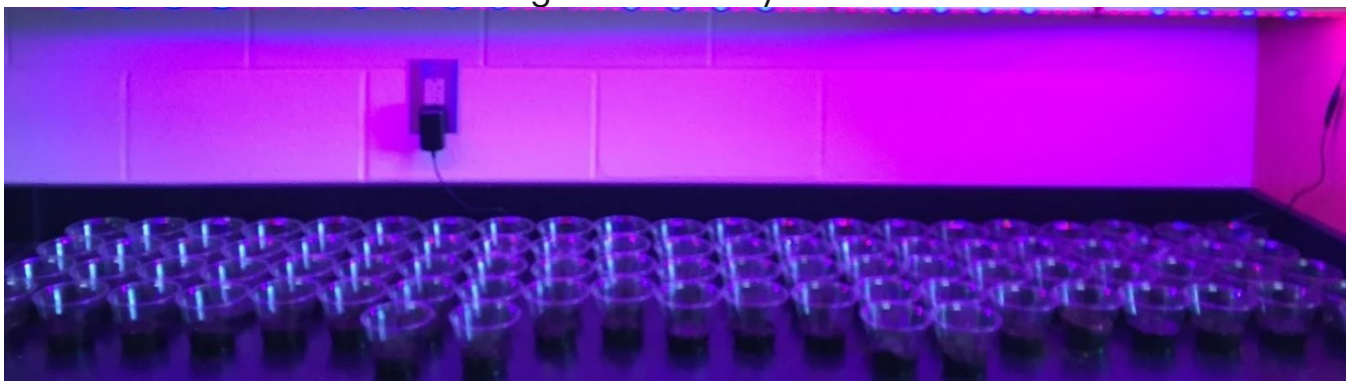
Spring Valley High School
August 2022 - May 2023



- Designed and built a small-scale test apparatus to evaluate whether a charged particulate filter could more efficiently remove soot from diesel particulate filters — using carbon black as a soot surrogate, steel wool electrodes, and a 3D-printed honeycomb DPF model
- Developed hands-on skills in soldering, circuit design, and 3D printing through end-to-end hardware fabrication; identified more realistic flow conditions as the key variable requiring future investigation
- Presented at SCJAS 2023, JSHS 2024, and UofSC Science & Engineering Fair 2023

The Effect of a Mixture of Magnesium Oxide and Titanium Dioxide Nanoparticles on Soil Health and the Growth of Vigna radiata

Spring Valley High School
August 2021 - May 2022



- Designed a controlled 3-week exposure experiment assessing nanoparticle mixture toxicity across two concentration levels, measuring soil pH, salinity, and plant height as outcome variables
- Found a consistent adverse effect — increased soil pH and reduced plant growth — across both exposure levels, contributing environmental impact data relevant to agricultural nanoparticle applications
- Presented at SCJAS 2022 and UofSC Science & Engineering Fair 2022; received Metric System Award at SCJAS 2022